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① Overview of Computing Paradigm:-

② A computing paradigm is a fundamental approach or style used to design and build computing systems. It defines how computation is performed where it happens, and how resources are used.

① Grid computing:-

② Grid computing is a computing paradigm in which multiple geographically distributed computers work together to solve large scale problems by sharing computing power, storage and data.

EX:- Weather prediction needs huge calculations. Instead of one computer, the work is shared among many comp so the result comes faster.

→ A Grid computing network mainly consists of three types of machines:-

- (i) Control Node.
- (ii) Provider.
- (iii) User.



→ Advantages

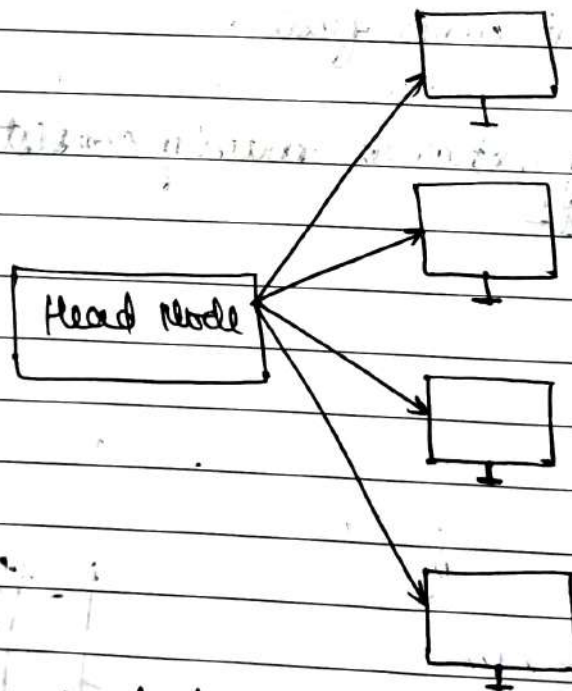
1. Faster processing.
2. Better use of Resource.

→ Disadvantage

1. Needs good Network.
2. Security Issues.

② Cluster Computing:-

- ① cluster computing is a system where computers (nodes) are connected in the same location and work together as one single system.



∴ cluster computing

- computers are in same place.
- High-speed LAN operation / connection.
- Appears as one system.

Ex:- In a college lab, many computers are connected to perform one heavy task like data analysis faster.

→ Advantages:-

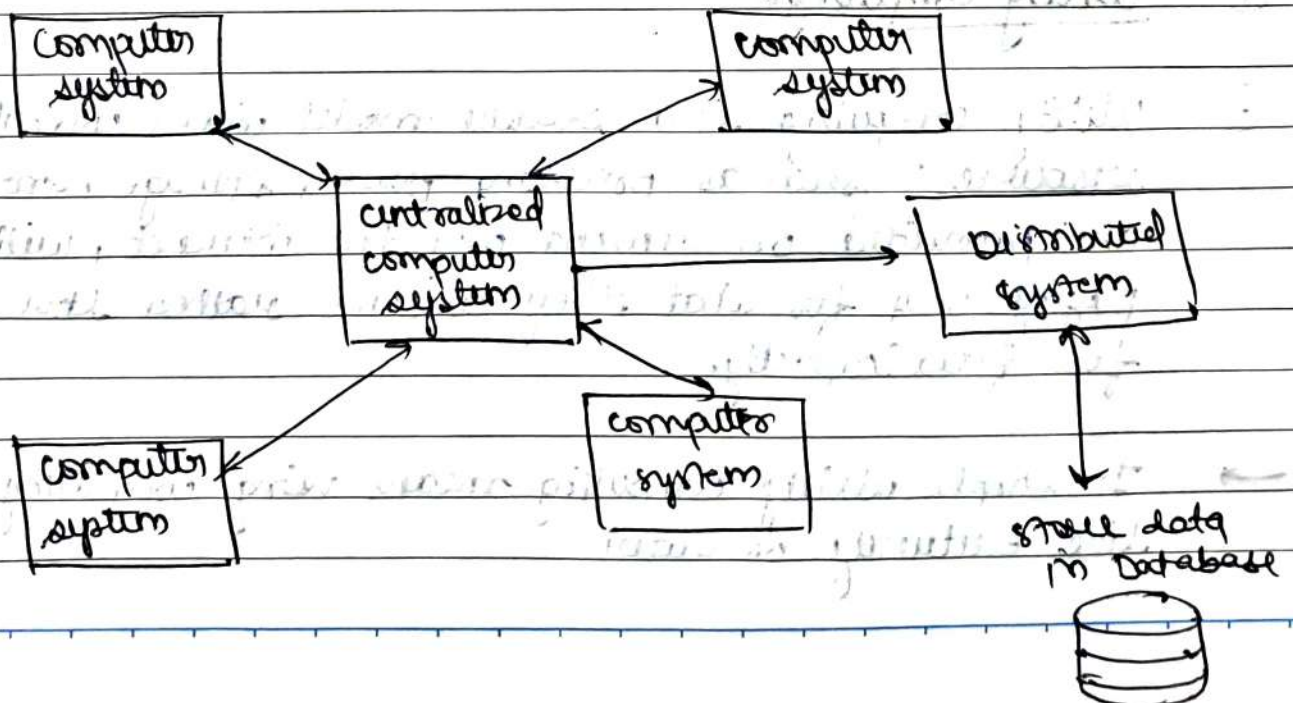
1. High speed.
2. Reliable.

→ Disadvantage:-

1. Expensive setup.
2. Needs maintenance.

③ Distributed computing:-

① Distributed computing is a system where multiple computers located at different places work together to solve a problem. Each computer has its own memory and processor, but they communicate over a network.



Ex- online banking system:- servers in different cities handle request together.

→ Characteristics

1. consists of multiple independent computers.
2. Nodes communicate and share the data through a network.
3. Appears as a single system to the end user.
4. Failure of one node does not stop the entire system.

→ Advantages

1. Faster processing
2. Fault tolerant.

→ Disadvantages

1. Network failure affects system.
2. Complex management.

④ Utility Computing

- ① utility computing is a service model where computing resources: such as processing power, storage, and software are provided on-demand over the internet, with users paying only for what they consume rather than paying for fixed capacity.

→ In simple utility computing means using computing resources like electricity or water.

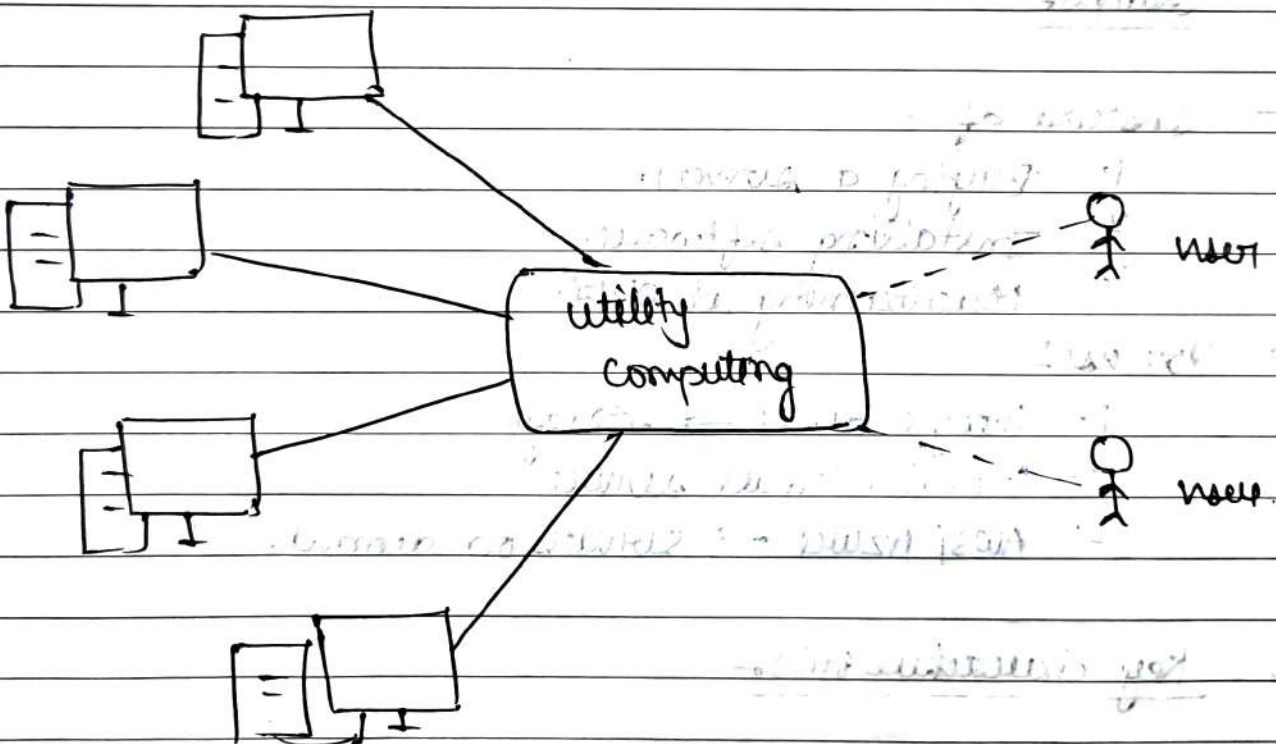
- Ex:-
- using cloud storage → pay for GB used,
 - using AWS/Azure → pay per hour.

→ Just like

- Electricity bill = units used
- utility computing = resources used

→ characteristics

1. Pay-as-you-use.
2. No need to buy servers.
3. Internet based.



→ Advantages

1. Pay only for what you use.
2. Scalable.

→ Disadvantages

1. Depends on internet connection.
2. cost may increase with heavy usage.

① cloud Computing

- ① cloud computing means using computing resources like (servers, storage, database, networking, software) over the internet instead of owning and maintaining physical hardware.

→ In short! Rent computing power instead of buying computer.

→ Example

→ Instead of :-

1. Buying a server.
2. Installing software.
3. Maintaining it 24/7.

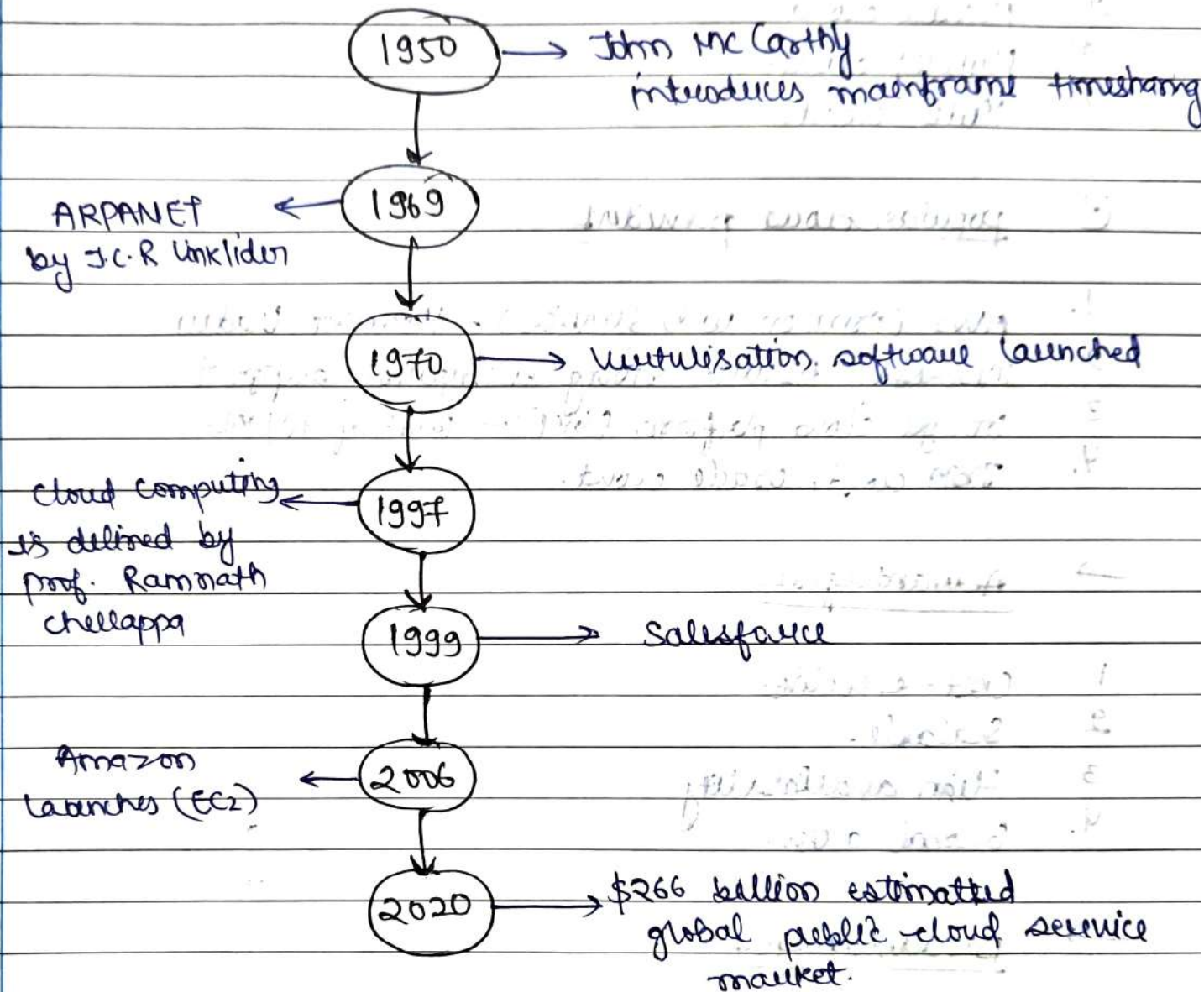
→ You use!

1. Google drive → storage
2. Gmail - email service.
3. AWS/Azure → Servers on demand.

① Key characteristics

1. on-demand self service.
2. Broad Network Access.
3. Resource pooling.
4. Rapid Elasticity.
5. Measured Service.

① Evolution of cloud computing



② cloud service models

1. IaaS → Infrastructure as a service.
2. PaaS → Platform as a Services.
3. SaaS → Software as a service.

① cloud Deployment Models

1. Public cloud.
2. Private cloud.
3. Hybrid cloud.
4. Multi-cloud.

② popular cloud providers

1. AWS (Amazon web services) - Market leader
2. Microsoft Azure - strong enterprise support.
3. Google cloud platform (GCP) - Best of AI/ML.
4. IBM cloud, Oracle cloud.

→ Advantages

1. Cost-effective.
2. Scalable.
3. High availability.
4. Global access.

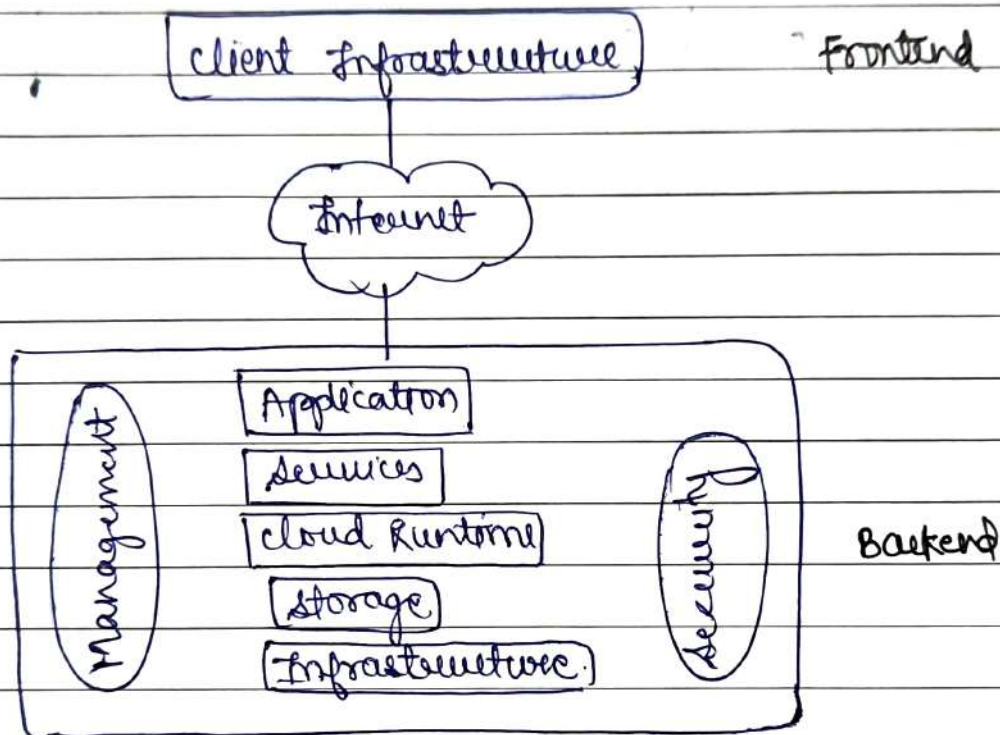
→ Disadvantages

1. Internet dependency.
2. Security risks.
3. Vendor lock-in.
4. Limited control in SaaS.

① NIST Model:-

→ The cloud architecture is divided into 2 parts

1. Frontend
2. Backend
3. A network (internet, intranet, or intercloud).



→ NIST:- National Institute of standards and technology (USA)

① cloud stakeholder as per NIST:-

→ cloud reference model as per NIST.

1. cloud provider.
2. cloud carrier.
3. cloud Broker.
4. cloud Auditor.
5. cloud consumer.



① Requisites / Benefits:-

① Scalability:-

1. Horizontal and Vertical scaling.
2. Automatic scaling.

② Global Reach:-

1. Worldwide Availability.
2. Multi-Region Deployment.

③ Disaster Recovery and Business continuity:-

1. Automated Backups.
2. Cross-Region Redundancy.

④ Security:-

1. Advanced security Features.
2. Shared responsibility model.

⑤ Role of open standards:-

1. Interoperability (compatibility).
2. Avoid vendor lock-in.
3. Portability.
4. Security & Trust.
5. Cost Reduction.
6. Better collaboration.
7. Innovation & Growth.

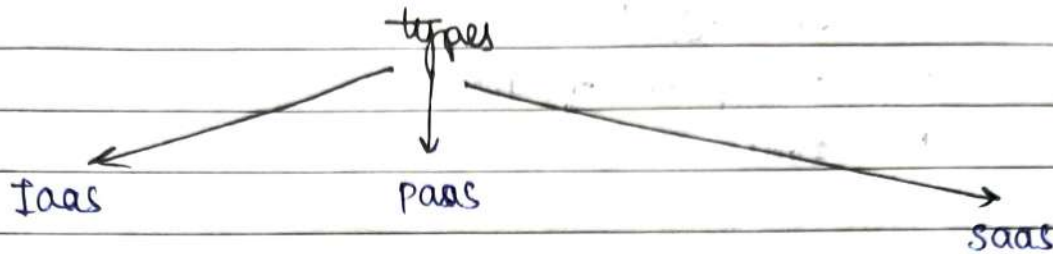
Ex:- (i) REST API.

(ii) openstack.

(iii) kubernetes. etc.

① Cloud computing Architecture

① cloud computing stack



1. IaaS!

- Infrastructure as a Service.
- Virtualized computing resources over the internet, including virtual machines, storage, networks, and load balancers. users have control over:-

1. operating system.
2. storage.
3. deployed application, while the cloud provider manages the underlying hardware.

Ex:- AMAZON EC2, Google compute Engine.

• Use case

- (i) Hosting websites.
- (ii) application development environments.
- (iii) big data processing.
- (iv) backup/recovery solutions.

2. PaaS

- platform as a services.

→ A platform for building, testing and deploying applications without worrying about underlying infrastructure

Include:-

- (i) operating system.
- (ii) databases
- (iii) development frameworks
- (iv) tools.

① usecase

① Rapid development and deployment of:-

- (i) web applications
- (ii) microservices.
- (iii) APIs.

Ex- Google App Engine, Azure app services.

③ SaaS:-

→ Software as a service.

→ It is a cloud service model where:

- The software is hosted on the cloud.
- Users access it through a browser or app.
- No need to install or maintain it manually.

Ex- Google Docs, Slack, Canva etc.

① usecases:-

- (i) Email services.
- (ii) Customer relationship management.
- (iii) E-commerce platform. etc.

① Role of Network in cloud computing:-

① The network plays a very important role in cloud computing because cloud services are accessed through the Internet.

② Access to cloud services:-

① Users can use cloud services like storage, applications and servers only through a network connection.

Ex:- Google Drive, AWS, Azure.

② Data Transfer:-

① Networks help in transferring data between the user and cloud services.

(i) uploading files.

(ii) downloading data.

(iii) sharing information.

③ Communication Between cloud components.

④ Resource sharing.

⑤ Scalability and Load Balancing.

⑥ Security and Reliable connectivity.

③ Protocol used:-

1. HTTP/HTTPS.

2. TCP/IP.

3. FTP/SFTP.

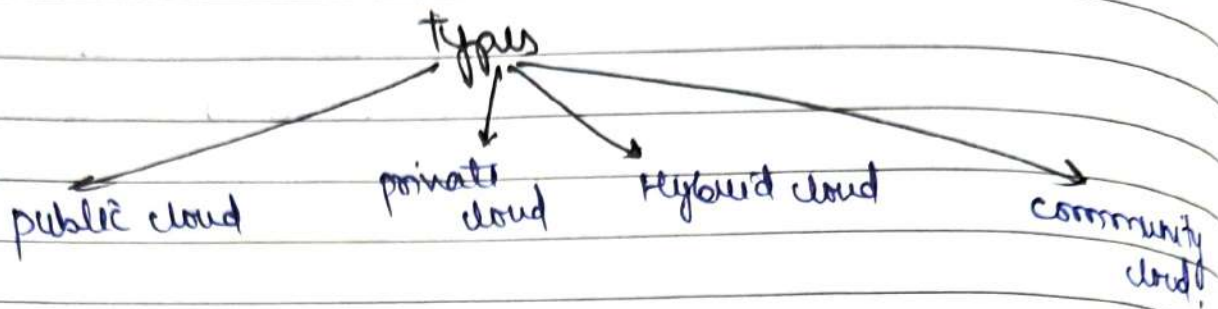
4. SMTP/IMAP/POP3.

5. DNS (Domain Name System).
6. VPN Protocols (IPSec, SSL).
7. SOAP and REST APIs.
8. SSH.

① Role of web server

1. Hosting websites and Web Application.
2. Handling client Requests.
3. Providing cloud services Access.
4. Managing Data communication.
5. Supporting scalability.
6. Security and Authentication.

① Deployment models



① Public cloud:-

- ① cloud services are provided over the internet by third party providers like AWS, Azure, or google cloud. It is cost-effective and scalable but less secure.

② Features:-

1. Accessible through Internet.
2. Shared resources.
3. Pay-as-you-go model

③ Advantages

1. Low cost.
2. Highly scalable.
3. Easy to use.

④ Disadvantages

1. Less control.
2. Security concerns for sensitive data.

Ex:- AWS, Azure, GCP, etc.

② Private cloud:

① cloud Infrastructure is used only by one organization. It offers high security and control, but it is more expensive.

② Features

1. Single tenant environment.
2. High security.
3. Managed internally or by third party.

③ Advantage:-

1. More control.
2. Better security.
3. Suitable for sensitive applications.

④ Disadvantage

1. Expensive.
2. Maintenance responsibility

③ Hybrid cloud

① It is a combination of public and private cloud. It provides flexibility, better security, and scalability.

② Features

1. Flexible environment.
2. Data sharing between clouds possible.

① Advantages

1. Best of both worlds.
2. cost-effective + secure.
3. cross scalability

② Disadvantages

1. Complex management.
2. Data integration challenges.

Ex: E-commerce websites (payments private, product catalog public).

③ Community Cloud

- ① cloud infrastructure is shared by multiple organizations with similar requirements, such as university or health-care institutions.

④ Features

1. shared inf. structure among community members.
2. common security + policies.

⑤ Advantages

1. cost shared
2. Better compliance
3. collaboration-friendly.

① Disadvantage

1. Limited scalability.
2. Shared governance issues.

Ex- university, Research institutions, Healthcare networks.

① Infrastructure as a Service:-

→ IaaS

② Introduction to virtualization:-

① Virtualization is a technology that allows multiple virtual machines (VMs) to run on a single physical computer. It enables the sharing of hardware resources like CPU, RAM, and storage efficiently.

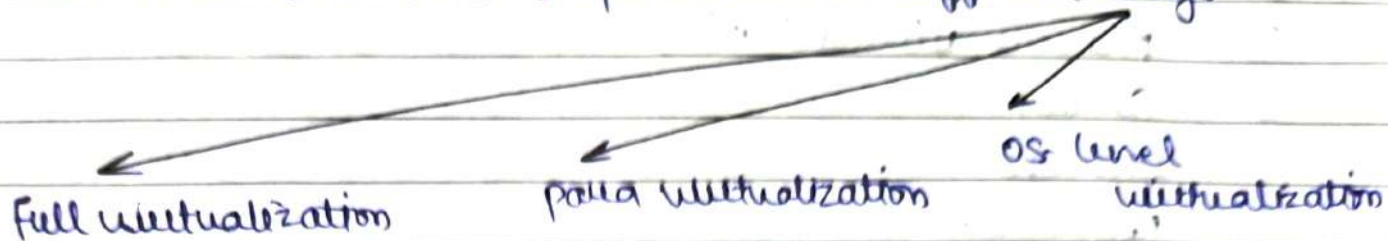
② A software called a hypervisor creates and manages the virtual machines.

→ Virtualization is widely used in cloud computing because

- (i) resource utilization (improve).
- (ii) reduce cost.
- (iii) provides scalability.

① Different approaches to virtualization:-

② Virtualization can be implemented in different ways:-



1. Full virtualization:-

② Complete virtual environment is provided, and guest OS runs without modification.

2. para virtualization:-

3. os level virtualization:-

1. Hypervisors.
2. Machine Image.
3. Virtual Machine (VM).
4. Resource virtualization.
5. Server.
6. Storage.
7. Network.
8. Virtual Machine (resource) provisioning and manageability.
9. storage as a service.
10. Data storage in cloud computing (storage as a service).

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① Platform as a service:- (PaaS)

② Introduction to PaaS:-

③ PaaS is a cloud computing services model that provides a platform and environment for developers to build, run and manage applications without managing the underlying infrastructure like servers, storage, and networking.

→ In simple:-

→ In PaaS, the cloud provider gives a complete platform that includes:

- (i) operating system.
- (ii) database.
- (iii) web server.
- (iv) development tools.
- (v) programming language environment.

→ Key Feature of PaaS:-

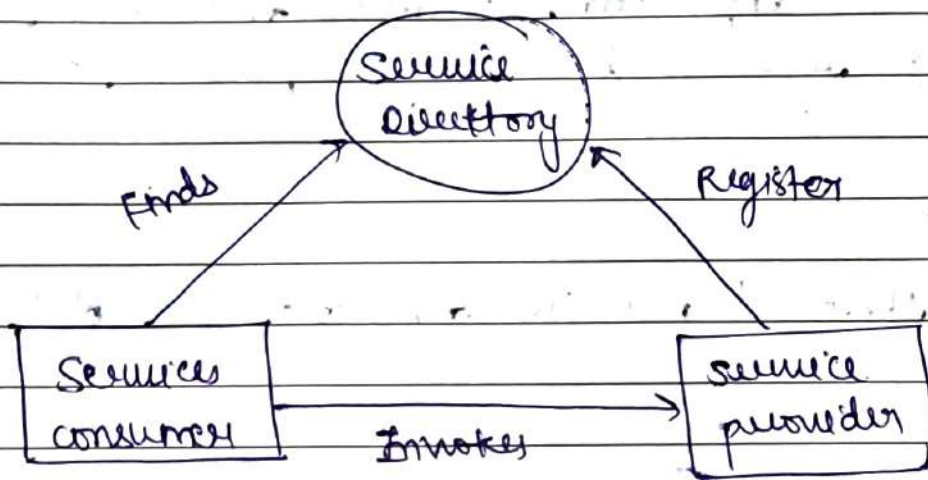
- 1. Scalability.
- 2. Cost effective.
- 3. Easy application development.

→ Ex. of PaaS:-

- 1. Google App Engine.
- 2. Microsoft Azure App Services.

① Service oriented Architecture:-

- ② SOA (Service oriented Architecture) is a software design approach where different functions of an application are broken into small, independent services. These services communicate with each other over a network.



③ Advantages

1. Reusability of services.
2. Easy maintenance.
3. Flexibility & scalability.
4. platform independent.

④ Disadvantages

1. High cost.
2. security issue.
3. complex to manage.

Ex:-

An E-commerce system:-

- (i) user service → handles login/registration.
- (ii) product services → manages product.
- (iii) order services → processes order.
- (iv) payment service → handles payments.

① SOA in Azure

② Microsoft Azure provides many tools and services to implement SOA-based application in the cloud.

③ Azure services used in SOA

1. Azure API management.
2. Azure Service Bus.
3. Azure Functions.
4. Azure Login Apps.
5. Azure Kubernetes Service (AKS).



① Software as a service! - (SaaS)

① Software as a service (SaaS) is a cloud computing model in which software applications are delivered over the internet instead of being installed on a local computer.

→ user can access the software through a web browser.

Ex- (i) Google Docs.

(ii) Microsoft 365.

(iii) Salesforce.

① characteristics of SaaS

1. Accessible through internet browser.
2. No installation required.
3. Subscription based payment.
4. Automatic updates.
5. Accessible from any device.

① Advantages

1. Low cost.
2. Easy maintenance.
3. Accessible anywhere.
4. Automatic upgrades.

① Disadvantages

1. Internet dependency.
2. Data security concerns.
3. Limited customization.

① Web Services

② Web services are software systems that allow different applications to communicate with each other over the internet.

→ They use standard protocols such as:

(i) HTTP.

(ii) XML.

(iii) JSON.

→ Types of web services

1. SOAP → Simple Object Access protocol.

2. REST → Representational state transfer. (CRUD).

③ Features

1. Platform independent.

2. Language independent.

3. Enables system integration.

Ex- A payment gateway API used by an e-commerce website.

④ Web 2.0:-

① Web 2.0 refers to the second generation of the web, where users can interact, share content, and collaborate online.

→ Earlier websites were static (web 1.0) but web 2.0 introduced dynamic and interactive websites.

① Features

1. user-generated content.
2. Social networking.
3. Collaboration.
4. Rich user interfaces.

Ex:-

1. Facebook.
2. YouTube.
3. Twitter.

① webOS

① web OS (web operating system) is an operating system that runs inside a web browser and allows users to access applications and files online.

→ Instead of storing data locally, everything is stored on cloud servers.

① Features

1. Access files from anywhere.
2. cloud storage.
3. online applications.
4. Platform independent.

Ex:-

1. Chrome OS.
2. eye OS.

① Service Management in cloud computing:-

② Service Management in cloud computing refers to the process of managing and delivering cloud services efficiently to users.

→ It ensures that cloud services are reliable, secure, and available according to the agreed standards between the provider and the customer.

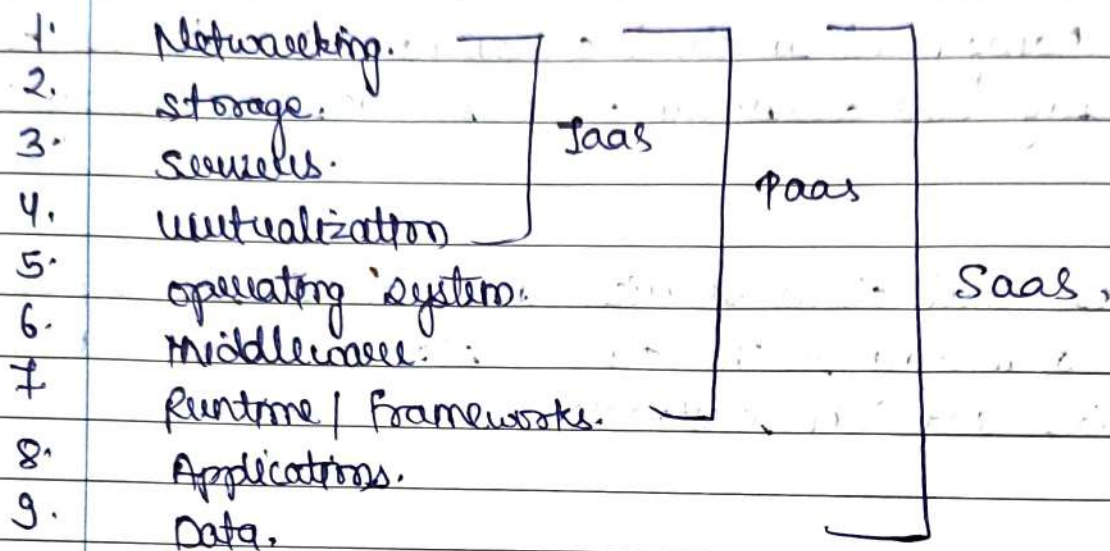
③ Service Level Agreement (SLA):-

④ A service level Agreement (SLA) is a formal contract b/w a cloud service provider and a customer that defines the level of service expected.

⑤ It specifies

1. performance standards
2. Availability
3. Responsibilities of both parties.
4. Penalties if the service level is not met.

→ 9 layers in cloud:-



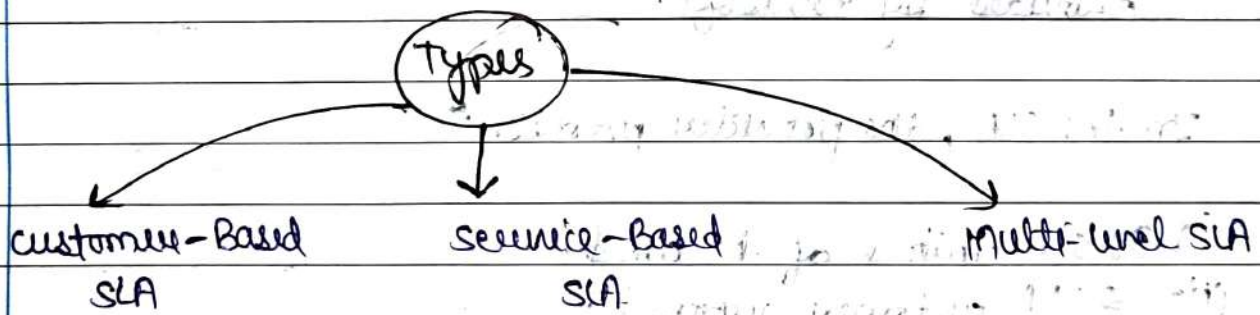
→ All these are called cloud.

① Service Level Agreement (SLA):-

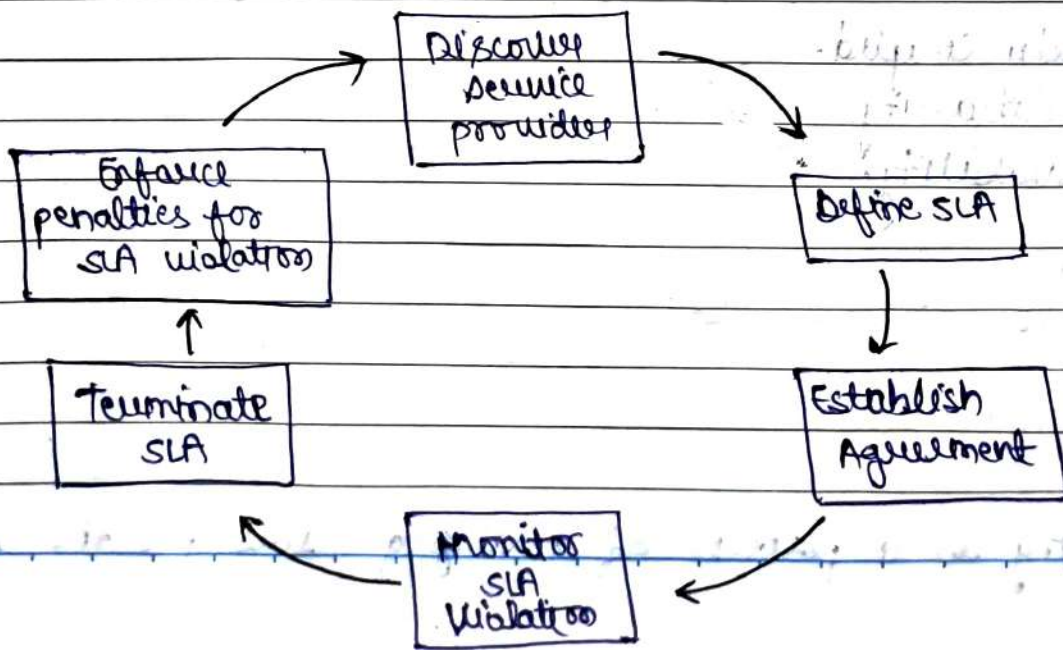
② A service level Agreement (SLA) is a formal contract between a service provider and a customer that defines the expected level of services

③ Why is SLA used:-

1. clarity of Expectations
2. Reliability Assurance
3. Accountability
4. Customer Trust
5. Business continuity



→ SLA Lifecycle:-



① Advantage:-

1. Improve service reliability.
2. Defines clear expectations.
3. provides accountability.
4. Helps monitor service performance.

① Disadvantages

1. Complex to create.
2. Hard to measure.
3. difficult to enforce penalties.

Ex:- Suppose a company uses cloud services from Microsoft Azure provided by Microsoft.

In the SLA, the provider promises:

- (i) 99.9% uptime of the cloud.
- (ii) 24/7 customer support.
- (iii) problems will be fixed within 2 hours.

→ loosely coupled.
→ modularity.
→ resiliability.

① Billing and Accounting

* Billing

① Billing is a process of charging users for the cloud.

service they use.

→ Billing Features

1. Pay-as-you-go pricing.
2. Subscription plans.
3. Cost estimation.
4. Automatic billing.

Ex:- (i) you use a virtual machine for 5 hours,
The system track 5 hours usage.

* Accounting

① Accounting is the process of tracking and recording the usage of cloud resource.

- Ex
- (i) How much storage is used.
 - (ii) How many hours servers are used.

→ Accounting Features

1. Usage tracking.
2. Resource monitoring.
3. Cost analysis.
4. Report generation.

⑥ Comparing Scaling Hardware Traditional vs cloud

① Traditional Scaling

→ on-premise.

- ① Require buying physical servers and hardware.
- ② Scaling takes time (days/week).
- ③ High initial cost (CapEx).
- ④ Limited flexibility.
- ⑤ Risk of underutilization or overload.

⑦ Traditional scaling is the process of increasing system capacity by adding physical hardware like servers and storage manually.

Ex:- A company buys 10 servers for future demand, but uses only 5 → waste of money.

② Cloud Scaling

① cloud scaling is the process of increasing or decreasing resources instantly over the internet based on demand using cloud platform like Microsoft Azure, AWS, etc.

- ① Scaling is instant (minutes).
- ② No upfront cost (pay-as-you-go).
- ③ Highly flexible and elastic.
- ④ Auto-scaling based on demand.

Ex:- During high traffic, servers increase automatically - no downtime.

① Economics of Scaling: Benefitting Enormously:-

① Economics of scaling means saving cost and gaining efficiency when scaling resources.

→ OR it means reducing cost and improving efficiency by scaling resource, especially in cloud where users pay only for what they use.

→ cloud Benefits Enormously:-

1. No upfront investment.
2. Pay only for use.
3. Auto-scaling.
4. Massive infrastructure.
5. Reduce maintenance cost.

Ext E-commerce

→ After sale, traffic decreases, extra servers removed. So, company pay less because fewer resources are used.

① Managing Data:-

② Looking at Data:-

③ Looking at Data means examining and understanding data before using it.

→ It helps to know:-

1. What type of data we have.
2. Whether data is correct or not.
3. How data can be used.

→ Important

1. Helps in better decision making.
2. Identifies errors or missing values.
3. Improve data quality.
4. Make data ready for analysis.

→ What we check while looking at Data:-

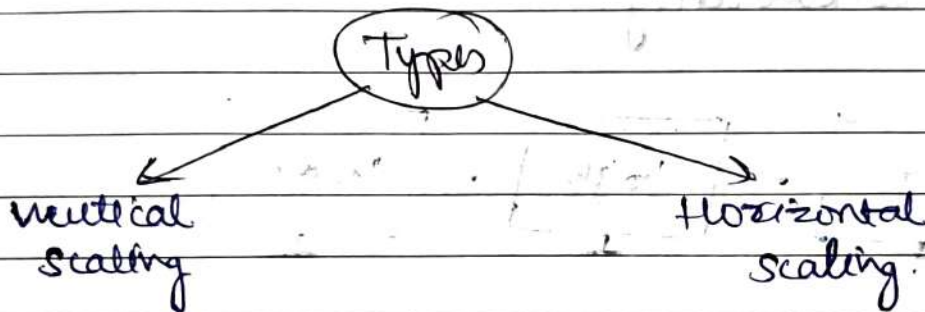
1. Data Type.
2. Data Structure.
3. Missing data.
4. Duplicate data.
5. Data Accuracy.

① Scalability & cloud services

1. Scalability:-

① Scalability is the ability of a system to increase or decrease resource according to demand.

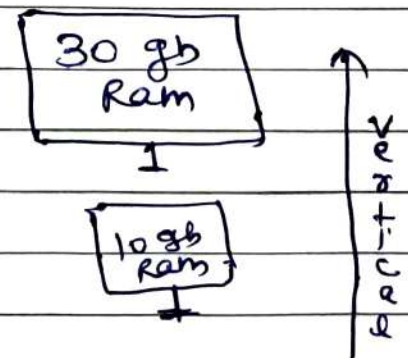
→ cloud platform like :- Azure, - AWS etc provide easy scalability.



1. Vertical scaling

① Increase power of a single machine, by adding more computing power like CPU and RAM to an existing machine.

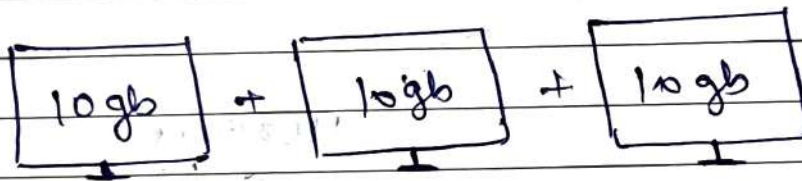
- Easy to implement.
- Data sharing is easy.
- It is cheaper.
- also known as scale up.
- Upgradation is easy.
- less concurrency.
- Consume less power.



2. Horizontal Scaling

① Horizontal scaling means we scale by adding additional machines to our existing bunch of resources.

- Difficult to implement.
- Data sharing is complex.
- Costly.
- More easy to upgrade in the future.
- Consumes more power.
- More concurrency.



→
Horizontal.

① Database & Data stored in cloud:-

① A cloud database is a database that is hosted and managed on cloud platform instead of local systems.

Ex:- AWS, Azure, etc.

→ photos in your phone gallery → local storage.
→ photos in Google Drive → cloud storage.

② Types of Data Stores in cloud:-

① Relational Databases (SQL):-

• Structured data

Ex:- MySQL, SQL Server.

② NoSQL Database:-

• Unstructured or semi-structured data.

Types:- Key-value, Document, Graph.

③ Object storage:-

• Stores files like image, videos.

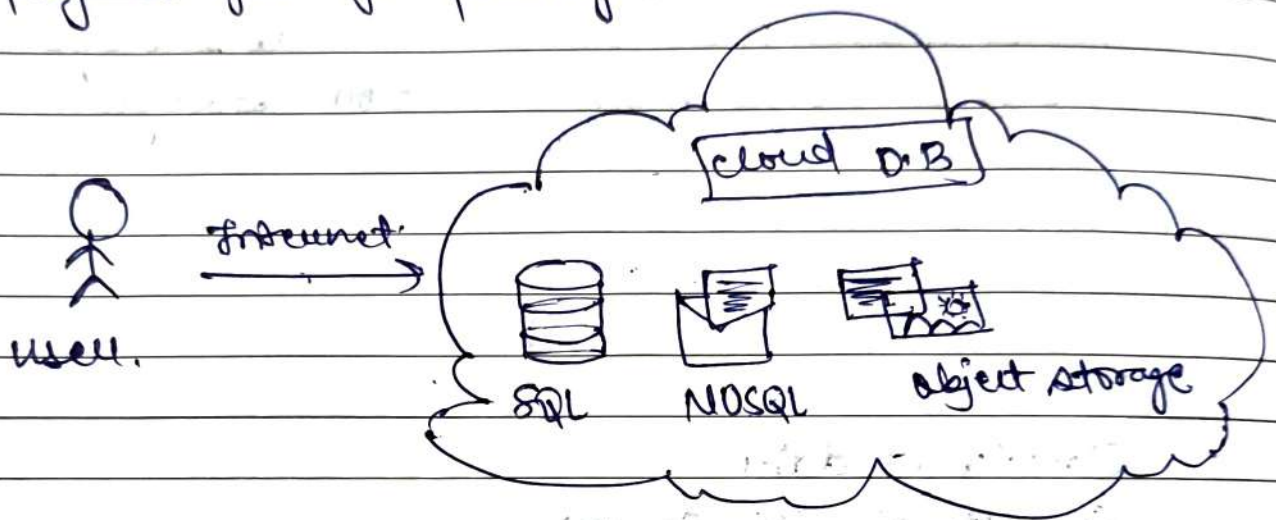
Ex:- cloud storage buckets.

④ Data Warehouses:-

• Stores large amounts of data for analysis.

① Why use cloud Data Storage?

1. Access from anywhere.
2. No need to manage hardware.
3. Safe backup.
4. Easy to increase storage.
5. pay-as-you-go pricing.



cloud Database.

① Large scale data Processing:-

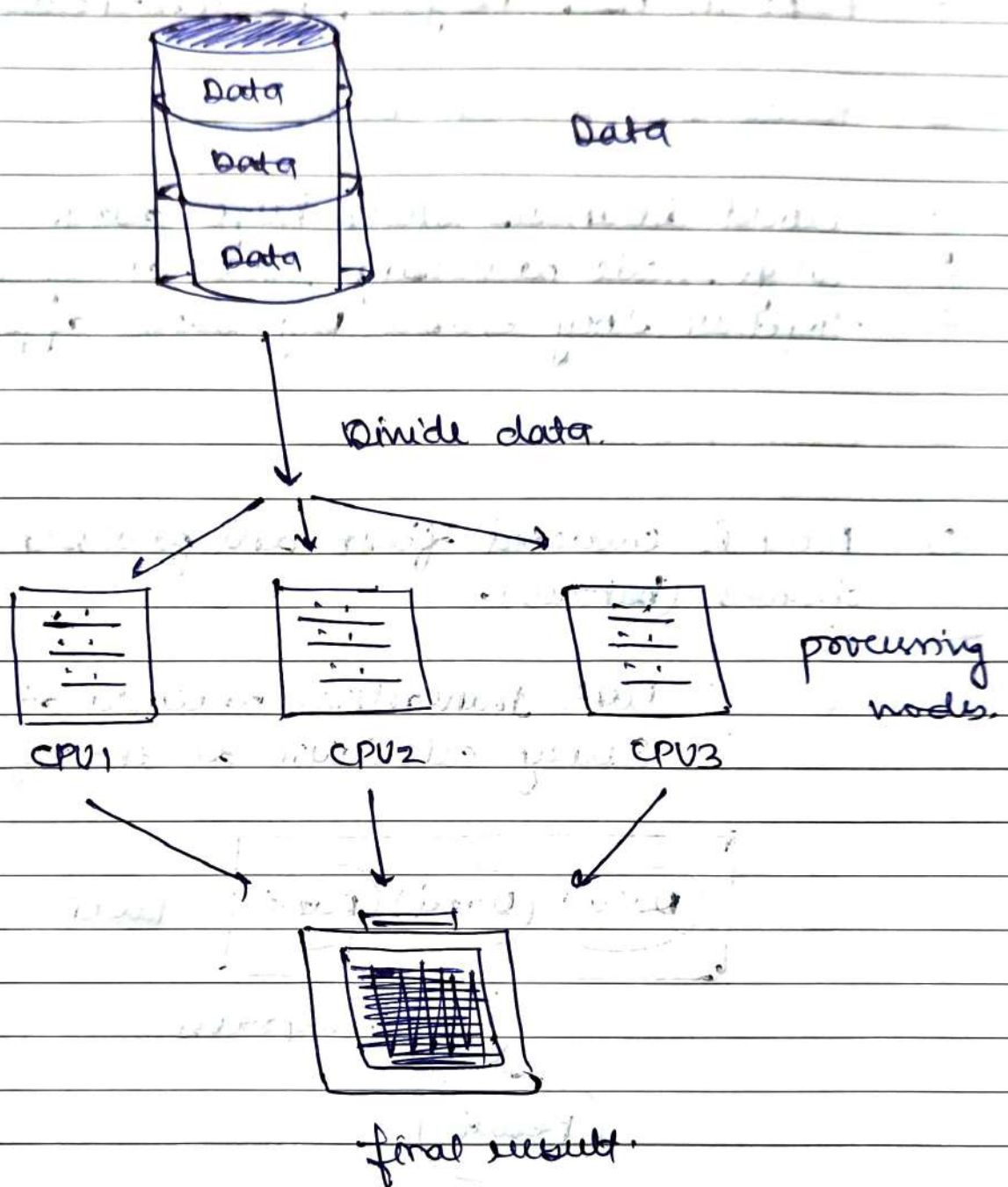
- ① It means handling and analyzing very large amount of data (Big data).

Example:-

- (i) Google processes billions of searches daily.
- (ii) YouTube processes huge video data.

This is a large scale data processing.

① How it works:-



Step 1:- Data is divided into small parts.
 Step 2:- processed on many computers on same time.
 Step 3:- Result are combined.

① Why it is needed:-

1. Fast processing of data.
2. Real time results.

3. Helps in AI, analytics, recommendations.

② Relationship Between both:-

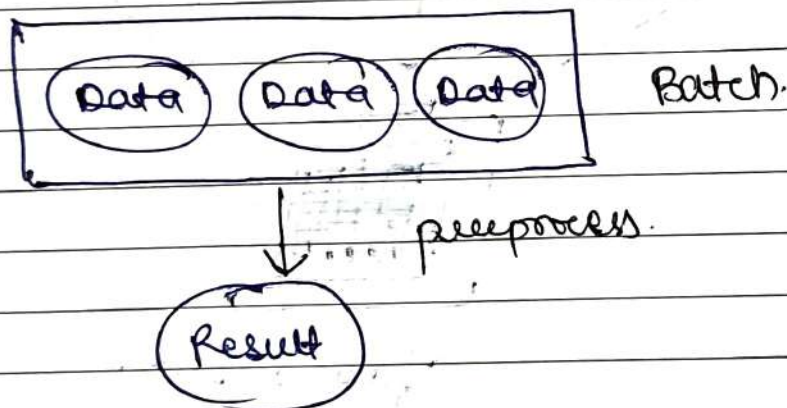
1. cloud databases store huge data.
2. large-scale processing tools analyze that data.
3. Together they enable Big Data applications.

③ Batch processing:-

③ Data is collected first and processed later in large chunks (batches).

→ slow.

- Ex:- (i) Bank transactions processed at night.
(ii) Salary calculation at end of month.



④ Stream processing:-

④ Data is processed immediately as it arrives (real-time).

Ex:- (i) live stock market update.

(ii) Instagram notifications.

(iii) Fraud detection.



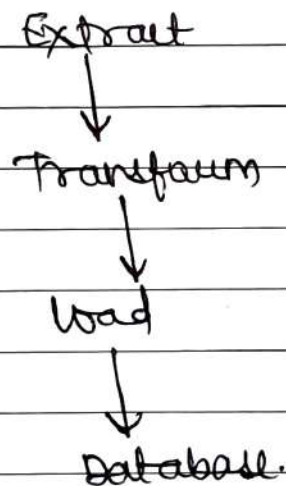
→ Fast.

⑥ ETL:-

⑥ Extract → Transform → Load

→ Data is cleaned and transformed before loading into database.

Flow:-



→ Storage.

→ Use:- Traditional system.

⑥ ELT:-

⑥ Extract → Load → Transform.

→ Data is loaded first, then transform inside the database.

Extract



Load



Transform



Database



Fastly



cloud systems (usi)

① Cloud security:-

② Infrastructure security:-

③ Infrastructure security means protecting the cloud system (servers, networks, data centres) from attacks.

Ex:- Azure, AWS etc.

④ Key security Aspects

1. Network security.
2. Physical security.
3. Virtualization security.
4. Storage security.
5. Identity & control Access.

⑤ How cloud providers secure infrastructure

1. Data encryption.
2. Regular security updates.
3. Monitoring & threat detection.
4. Backup & disaster recovery.

⑥ Importance:-

1. Protects sensitive data.
2. Prevents cyber attacks.
3. Ensures system availability.
4. Builds user trust.

Example using Microsoft Azure.

- (i) Data is stored securely.
- (ii) Access is controlled.
- (iii) System is monitored 24/7.

① Network level security:-

② Network level security means protecting the cloud network and data while it is being transferred over the internet from unauthorized access, attacks, or misuse.

→ cloud platform like:-

(i) Azure.

(ii) AWS, provides build-in network security features.

③ Key components:-

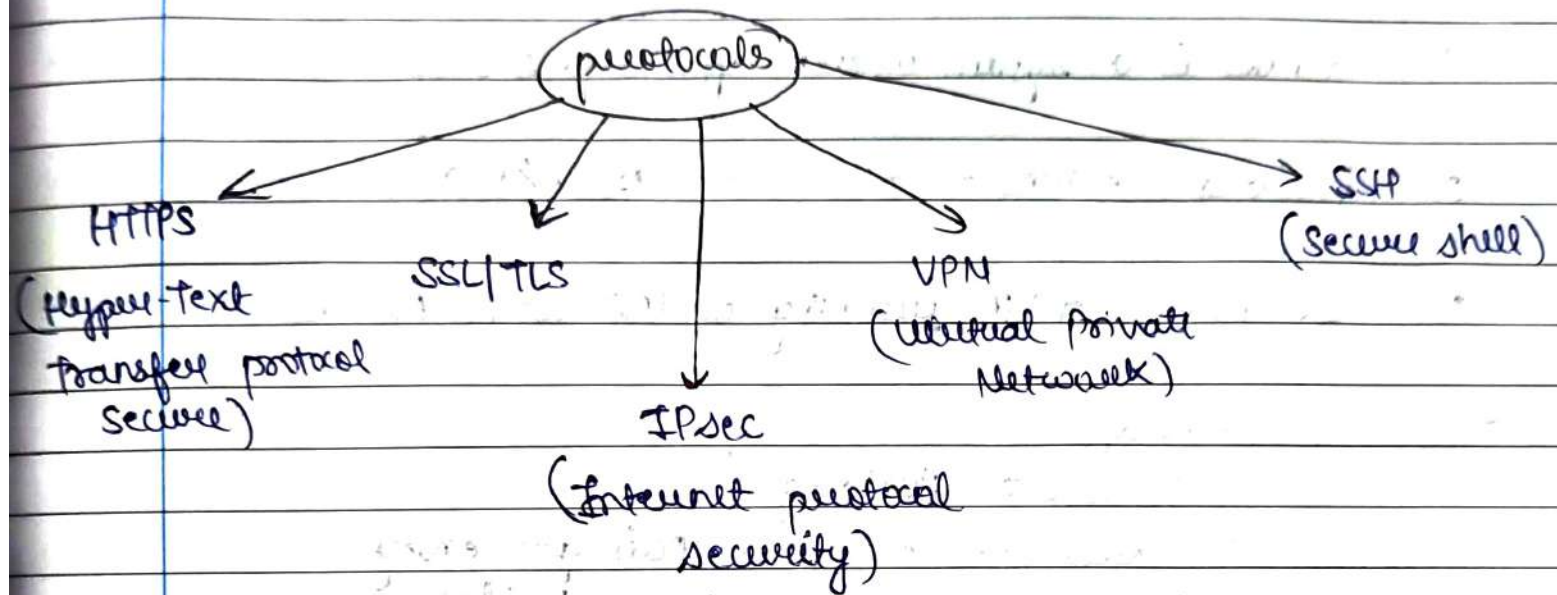
1. Firewalls. (control incoming and outgoing traffic).
2. Virtual private Network (VPN). (secure private connections).
3. Encryption. (convert data into secure code).
4. Intrusion Detection & Prevention (IDS / IPS).
5. Network segmentation. (For better security).

④ Why it is important

1. Protects data from hackers.
2. Ensures safe communication.
3. Prevents data theft.
4. Maintains privacy.

Example:- In Microsoft Azure

- (i) Firewall control traffic.
- (ii) VPN secures connection.
- (iii) Data is encrypted during transfer.



② Host Level Security:-

① Host level security means protecting individual machines (hosts) like virtual machines (VMs), servers, and operating systems in the cloud.

→ A host is a system where applications run!

① cloud → VMs → OS → Applications.

① Example - A VM running on Microsoft Azure.

→ Think of:

→ cloud = Building

→ Network security = Main gate entry

→ Host (VM) = Rooms inside building.

Even if the main gate is secure.

If a room is open → anyone can enter.

① What is done in Host level security?

1. Login Security.
2. OS updates.
3. Firewall.
4. Antivirus.

③ Application level security :-

- ① Application level security refers to protecting software applications from threats, attacks, and unauthorized access.

→ When a user interacts with an application:



→ Application security ensures:

- (i) Only valid users can access.
- (ii) Input is safe.
- (iii) Data is protected.

④ Key security Mechanisms

1. Authentication:-

→ Verify who the user is.
Example:- username + password.

2. Authorization:-

→ Decides what the user can access.
Example:- Admin vs normal user.

3. Input validation:-

→ Check user input to prevent attacks.
prevents SQL injection, malicious data.

4. Encryption:-

→ protects sensitive data (passwords, payments).

5. Session Management:-

→ Manages user login sessions securely.

⑥ Example:-

→ Think of,

∴ Instagram / any app

Application security ensures:

- (i) only you can log in.
- (ii) others cannot access your account.
- (iii) your data is protected.

⑦ Why it is important:-

- (i) Protects user data.
- (ii) Prevents hacking.
- (iii) Ensures safe application usage.

① Data Security and Storage

② Data security means protecting data from unauthorized access, loss, or theft, especially when it is stored in the cloud.

③ cloud platforms like:-

(i) Azure

(ii) AWS etc. store large amounts of user data, so security is very important.

④ Data Storage

⑤ Data storage means saving data in cloud servers instead of local drive.

⑥ Data privacy and security Issues

⑦ Data privacy and security issues are problems related to protecting sensitive data from unauthorized access, misuse, or loss.

⑧ Data Breach

→ Unauthorized access to sensitive data.

Example:- Hackers stealing user data.

⑨ Data Loss

→ Data gets deleted due to errors or failure.

DOMS
Date / / Page No.

③ Unauthorized Access:-

→ Someone accesses your data without permission.
Example: Hacking your account.

④ Data Leakage:-

→ Sensitive data is exposed accidentally.
Example: private data shared publicly.

⑤ Lack of control:-

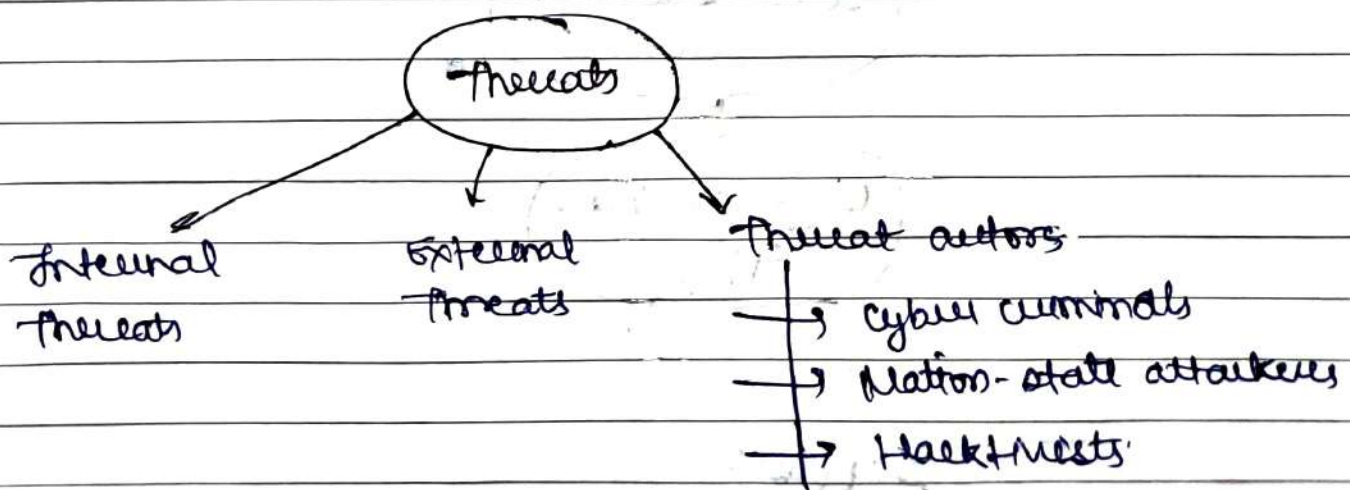
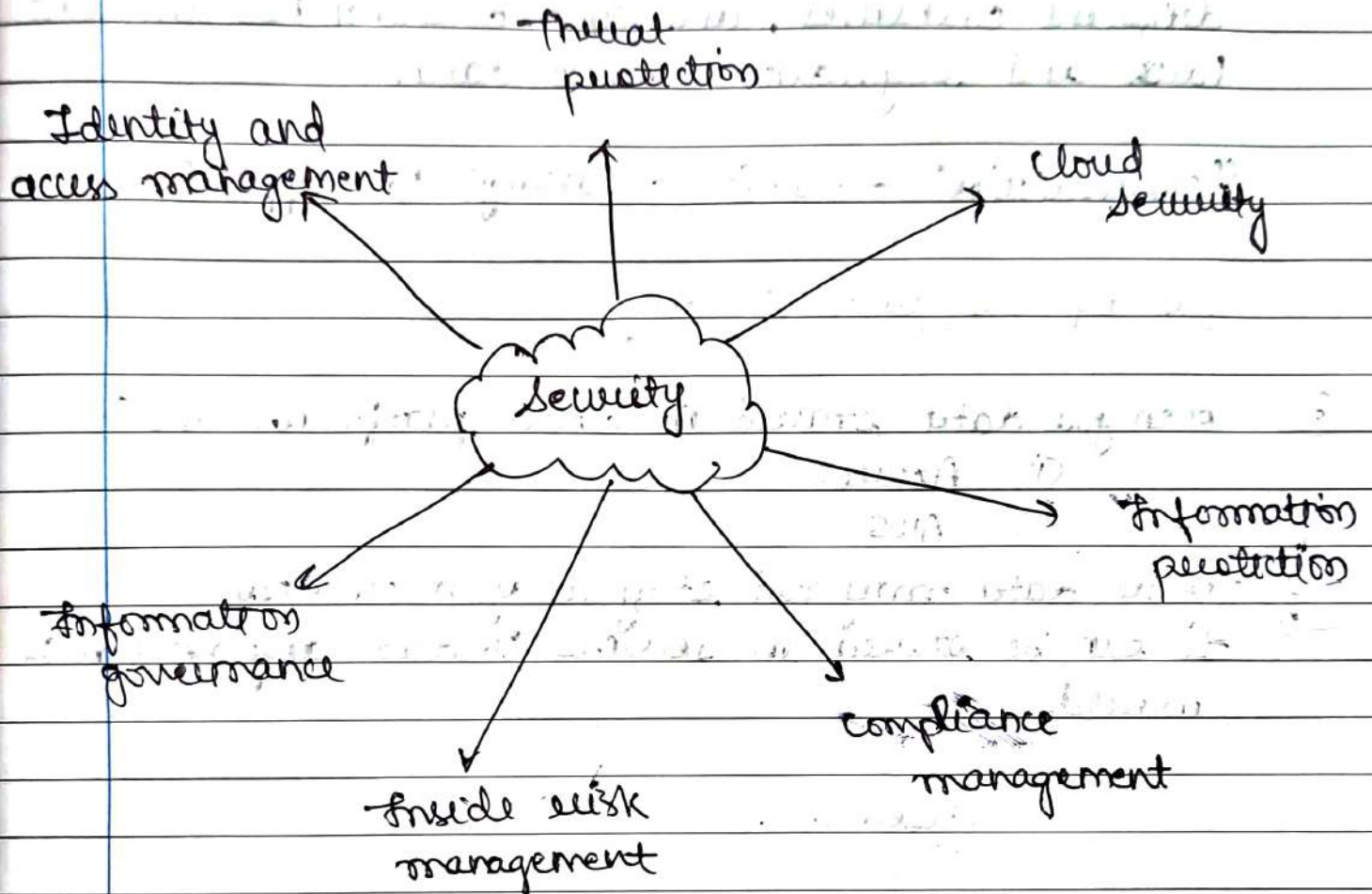
→ users don't know where their data is stored.

⑥ Inside Threats:-

→ Employees misuse data.
Example:- staff accessing user information illegally.

⑦ Weak security:-

→ poor protection systems make hacking easy.



① Jurisdictional Issues Raised by Data Location:-

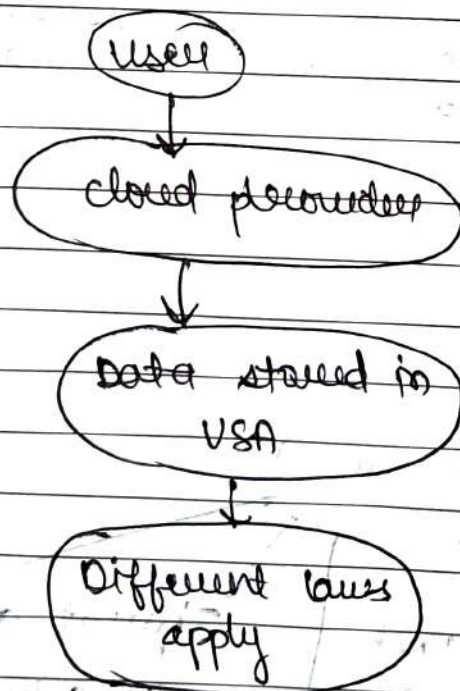
② Jurisdictional issues arise when data is stored in different countries, and each country has different laws and regulations about data.

"Jurisdiction" → which country's law applied.

③ step-by-step understanding:-

④ when your data stored in cloud platforms like:-
 (i) Azure.
 (ii) AWS.

→ your data may not stay in your country.
 → It can be stored in servers located anywhere in the world.



Example:-

Suppose:-

- you are in India
- using Azure
- your data is stored in USA

→ Now:

- Indian law say on thing
- US law says another.

Note - ① It can create the Big Issue.

② Key issues

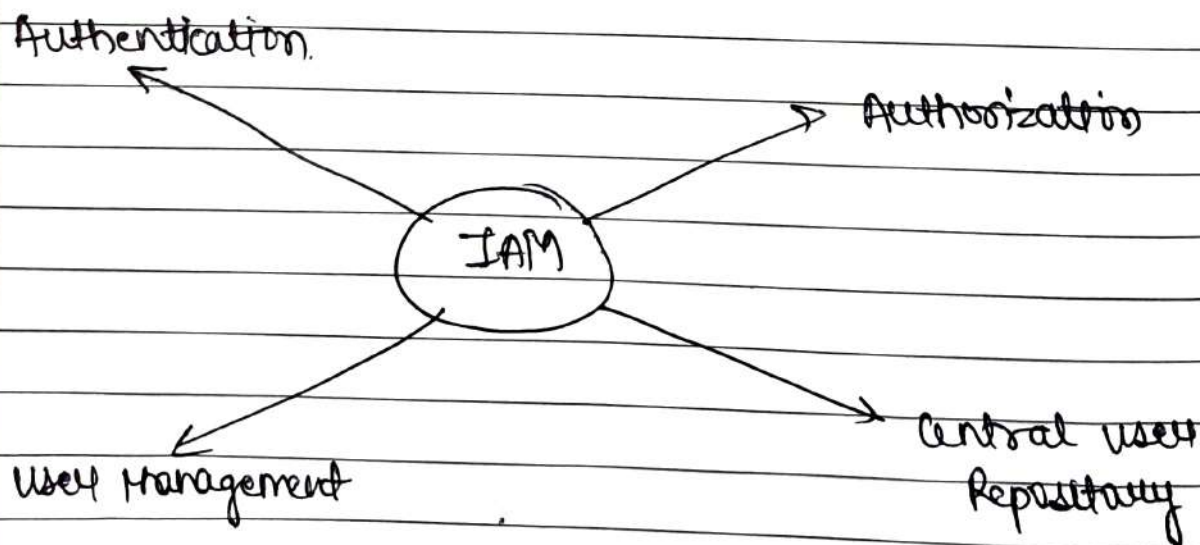
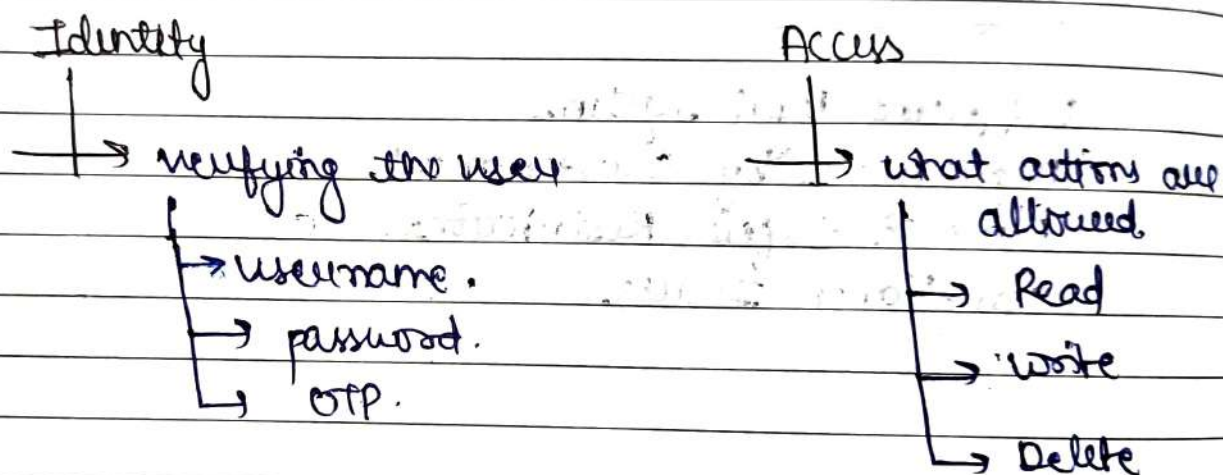
1. Different legal systems.
2. Data Access by Government.
3. Data Transfer Restrictions.
4. Compliance Issues.

① Identity & Access Management

② IAM is a security framework of policies, technologies, and processes ensuring authorized user (humans, devices or software) securely access the right resource at the right time.

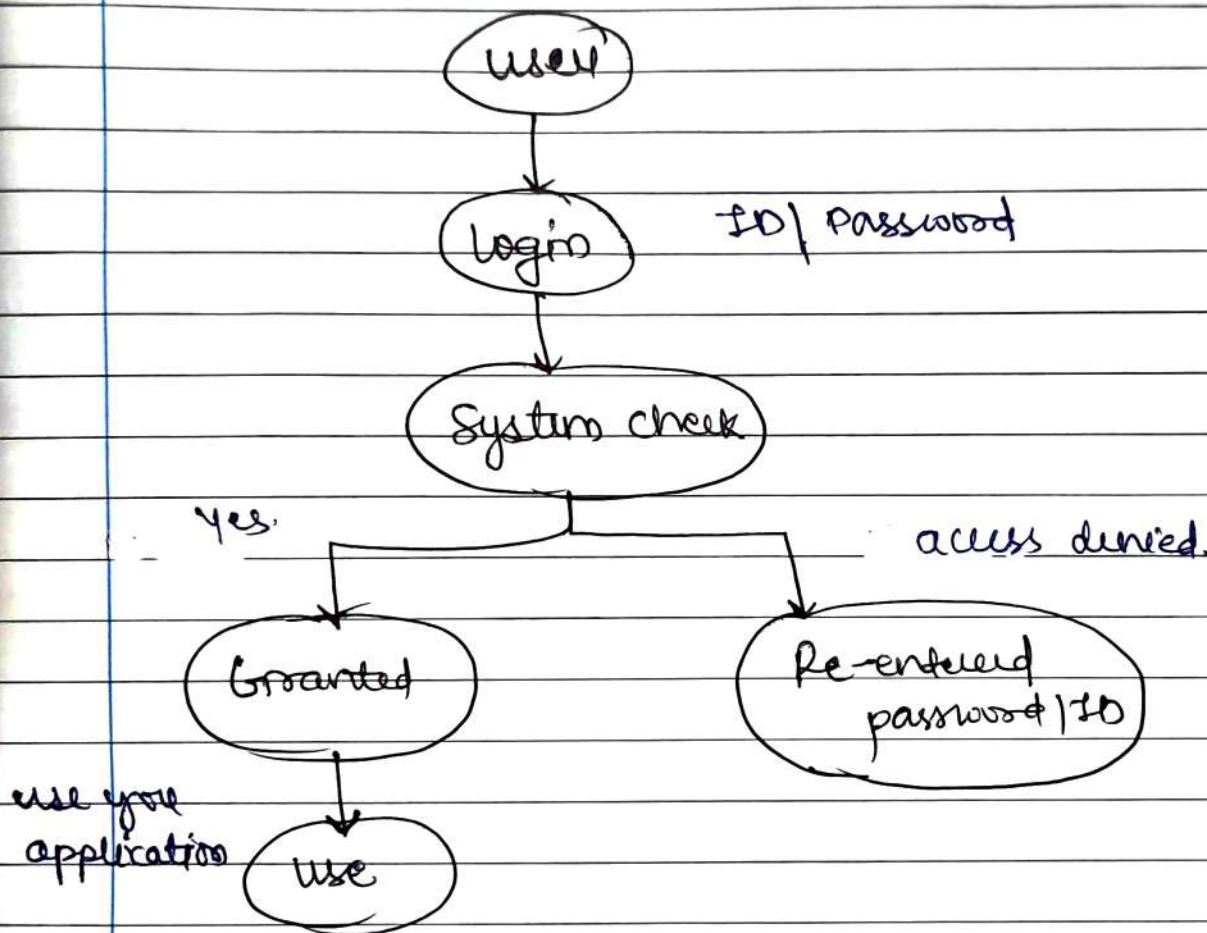
③ IAM is a system that controls:

- who are you (Identity).
- what you can do (Access).



① Why IAM is important

- (i) Prevent unauthorized access.
- (ii) Protect sensitive data.
- (iii) Ensures security and control.



Example

Think of a college system:

Step 1: login

→ you enter ID & password

→ system check ⇒ who are you.

(Authentication)

Step 2:- After login

- student → can view marks.
- teacher → can edit marks.

(Authorization)